



Semester 2 Examinations 2014/ 2015

Exam Code(s) 1MEC1, 1HDS1, 3BMS2, 3BS9, 4BME1, 4FM2
Exam(s) 3rd Science, 4th Science, 4th Financial Mathematics and Economics, 4th Mathematics and Education,

Module Code(s) MP491
Module(s) Nonlinear Systems

Paper No. 1
Repeat Paper No

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Internal Examiner(s) Dr P. Piironen

Instructions: For full marks answer question 1 in Section A **and** TWO questions in Section B correctly.
All questions carry the same weight.

Duration 2 hours
No. of Pages 5
Discipline(s) Applied Mathematics
Course Co-ordinator(s)

Requirements:

Release in Exam Venue	Yes	☒	No	☐
MCQ	Yes	☐	No	☒
Handout	None			
Statistical/ Log Tables	None			
Cambridge Tables	Yes			
Graph Paper	None			
Log Graph Paper	None			
Other Materials	Calculator			
Graphic material in colour	Yes	☐	No	☒

Section A - Answer all questions (a)-(d). (**5 Marks**/question)

1. (a) Consider the nonlinear ordinary differential equation

$$\frac{dx}{dt} = 2x^2 + 10x + 12.$$

Find all equilibrium points and determine if they are stable or not.

- (b) Consider the linear system of ordinary differential equations

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Classify the equilibrium point $(x, y) = (0, 0)$ and sketch the phase-plane portrait.

- (c) Consider the system of nonlinear ordinary differential equations

$$\begin{aligned} \frac{dx}{dt} &= y - e^{ax} - x^2, \\ \frac{dy}{dt} &= -y + x^2 + 1. \end{aligned}$$

Find the equilibrium point and determine for what values of a it is stable.

- (d) Consider the logistic map for $n = 0, 1, 2, \dots$ given by

$$x_{n+1} = 2x_n - 2(x_n)^2.$$

Find all fixed points and determine if they are stable or unstable.

Section B - Answer any two questions.

2. (a) Consider the one-parameter family of nonlinear ordinary differential equations

$$\frac{dx}{dt} = x(x + 2 + \mu)(\mu - x^2),$$

where μ is a real parameter.

- i. Find all equilibrium points.
- ii. Sketch the bifurcation diagram, and indicate the behaviour of non-equilibrium solutions using appropriate arrows.
- iii. Find all bifurcation points and classify them.

10 Marks

- (b) Consider the Selkov model of glycolysis given by the system of ordinary differential equations

$$\begin{aligned}\dot{x} &= -x + ay + x^2y, \\ \dot{y} &= 1 - ay - x^2y,\end{aligned}$$

where $a \neq -1$ is a real parameter.

- i. Describe how the position of the only equilibrium point varies with the parameter a .
- ii. Find the bifurcation point (x, y, a) for which a Hopf bifurcation occurs.
- iii. Describe how the nature of the equilibrium point changes at the Hopf bifurcation for varying a .

10 Marks

3. Consider the second-order nonlinear ordinary differential equation

$$\ddot{z} - z^2 + 4 = 0.$$

(a) Show that it can be written as

$$\begin{aligned}\dot{x} &= y, \\ \dot{y} &= x^2 - 4.\end{aligned}$$

2 Marks

(b) Show that the ODE system in part (a) is Hamiltonian, find the Hamiltonian function and show that

$$\frac{d}{dt}H(x, y) = 0.$$

6 Marks

(c) Use the Hamiltonian function found in part (b) to find and classify all equilibrium points.

4 Marks

(d) Use the Hamiltonian function found in part (b) to sketch the phase-plane portrait.

8 Marks

4. Consider the following discrete model, that describes the evolution of a population P , given by

$$P_{n+1} = \frac{rP_n}{1 + P_n},$$

where P_n is the population at year $n = 0, 1, 2, \dots$ and r is a reproduction rate parameter.

- (a) Find all fixed points and investigate for what values of r the model only has physical solution ($P_n \geq 0$).

5 Marks

- (b) Calculate the stabilities of the fixed points found in (a) when varying $r \in (-\infty, \infty)$.

5 Marks

- (c) Sketch the bifurcation diagram for $-3 \leq r \leq 3$.

5 Marks

- (d) Use the results in (a)-(c) and determine if the population P can undergo any bifurcations when $P > 0$. Explain your answer.

5 Marks